

Muhammad Abdullahi Said

Cape Town, South Africa



[Gmail](#)



[linkedin](#)



[Google Scholar](#)



[GitHub](#)

RESEARCH INTERESTS

Interests: AI Safety & Alignment, Explainable AI (XAI), Responsible AI, Large Language Models (LLMs), NLP. *Google DeepMind Scholar seeking to bridge theoretical mathematical foundations with scalable, real-world AI applications.*

EDUCATION

M.Sc. in Mathematical Sciences (AI for Science)

African Institute for Mathematical Sciences (AIMS)

Expected July 2026

Cape Town, South Africa

- **Scholarship:** **Google DeepMind Scholar** (Highly competitive full scholarship awarded to top African talent in AI).
- **Mentorship:** Mentored by Dr. Andrea Banino (Senior Research Scientist, Google DeepMind).
- **Key Coursework:** Monte Carlo Simulations, Bayesian Reasoning, LLM & NLP, CUDA Programming, Responsible AI, XAI.
- **Instructors:** Studied under Prof. Sanjoy Mahajan (MIT), Prof. Ulrich Paquet (Google DeepMind), Prof. Harrison Prosper (FSU), Prof. Mike Giles (Oxford)

M.Sc. in Big Data Technologies

University of East London

Date – 2024

London, United Kingdom

- **Grade:** Distinction.
- **Thesis:** Financial Fraud Detection: A Comparative Analysis of Machine Learning Approaches.
- **Supervisor:** Dr. Mahmoud Elbattah.

B.Eng. in Computer Engineering

Bayero University Kano

Date – 2019

Kano, Nigeria

- **Grade:** Second Class Upper.
- **Scholarship:** NNPC/CHEVRON National University Scholarship Award.
- **Project:** Door Access Control Using RFID.
- **Supervisor:** Dr. Sani Halliru.

SELECTED RESEARCH PROJECTS

Safe in the Future, Dangerous in the Past | *Dr. Shamsudden Muhammad*

Dec 2025

- **LLM Red-Teaming:** Conducted a systematic safety audit of SOTA models (GPT-5.1, Gemini 3, Claude 4.5) using **HausaSafety**, a custom adversarial dataset grounded in West African threat scenarios.
- **Experimental Design:** implemented a 2×4 factorial design to analyze **Cross-Lingual Alignment**; identified Temporal Asymmetry, where past-tense prompts bypassed safety guardrails significantly more often than future-tense.
- **Impact:** Challenged the multilingual safety gap narrative by discovering **Reverse Linguistic Gaps** in Claude 4.5; code and dataset released to facilitate non-western safety research.

Scalable Recommendation System via Alternating Least Squares (ALS) | *Prof. Ulrich Paquet*

Dec 2025

- **Objective:** Engineered a collaborative filtering engine on the **MovieLens 32M** dataset, focusing on computational efficiency.
- **Optimization:** Utilized **ALS** over Stochastic Gradient Descent and leveraged **Numba** for JIT compilation to parallelize matrix operations on a multi-core CPU.
- **Implementation:** Handled $>90\%$ data sparsity and implemented a "Cold Start" strategy using global average fallbacks.
- **Tech Stack:** Python, Numba, NumPy, Pandas.

Polarization Detection in Low-Resource Languages (LLMs) | *Dr. Shamsudden Muhammad*

Nov 2025

- **Objective:** Fine-tuning multilingual transformers (afro-xlmr) to detect sentiment polarization in **Hausa** (low-resource) and English.
- **Methodology:** Implementing **Cross-Lingual Transfer Learning** to mitigate data scarcity in Hausa.
- **Tech Stack:** PyTorch, Hugging Face Transformers, Weights & Biases.

PREPRINTS

Safe in the Future, Dangerous in the Past: Dissecting Temporal and Linguistic Vulnerabilities in LLMs
M. A. Said, M. S. Sani | arXiv preprint, 2025

WORK EXPERIENCE

Technical Research Assistant | AI Safety Nigeria Jan 2025 – Present

- Conducted research on AI safety, alignment failures, and jailbreak robustness in multilingual settings.
- Designed and evaluated red-teaming benchmark targeting paradigm-level safety blind spots.
- Facilitated fellowships and discussions on responsible AI safety practices.

CONFERENCES & WORKSHOPS

Quantum Machine Learning Workshop | AIMS South Africa Nov 2025

- Participated in technical sessions exploring quantum algorithms, variational quantum circuits, and their applications in modern machine learning tasks.

WAFIRA Capacity Building Workshop | AIMS South Africa Oct 2025

- Participated in the "Special Session for Male AIMS Students" focused on transformative gender equality norms and co-leadership strategies to support the advancement of women in African academia and research.

TECHNICAL SKILLS

Languages: Python, SQL
Frameworks & Libraries: PyTorch, Numba, Keras, PySpark, Scikit-learn, Hugging Face
Concepts: Bayesian Inference, Monte Carlo Methods
Tools: Git, Docker, Jupyter, Databricks, Tableau

PROFESSIONAL MEMBERSHIPS

AI Safety Nigeria: Member (2026 – Present)
International Society for Data Science and Analytics: Member (2025 – Present)
International Association of Engineers: Member (2021 – Present)
Nigerian Society of Engineers: Graduate Member (2021 – Present)

HONORS & AWARDS

Society for AI Advancement (2026): Africa’s 100 Rising AI Developers Initiative.
GoogleDeepMind Scholarship (2025): Full scholarship for MSc in AI for Science in AIMS.
ALX-T Data Analyst Scholarship (2022).
NNPC/CHEVRON National University Scholarship (2016): Merit-based award for academic excellence.
Junior Achievement Nigeria (2016).

CERTIFICATIONS

Artificial Intelligence Certification, Edureka (2024)
Data Science with Python, Edureka (2024)
Data Analyst Nanodegree, Udacity (2022)
Google Data Analytics Professional Certificate, Coursera (2022)